4.2

Technology and Production of Goods and Services

Cambridge IGCSE Business

- Automation & Mechanisation
- CAM
- 3D Printing
- Technology in Services
- Effects on Businesses
- Effects on Employees

Technology in Production

Technology is transforming how goods and services are produced. Businesses that adopt technology effectively gain significant competitive advantages.

- Distinguish automation from mechanisation
- Understand CAM and 3D printing with examples
- See how technology improves service sector productivity
- Evaluate advantages for businesses
- Evaluate disadvantages for employees

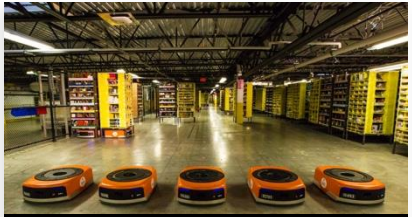
Automation & Mechanisation

Machines replacing or assisting human workers

Automation vs Mechanisation – Key Differences & Real Examples

Automation replaces human workers entirely with computers making decisions. Mechanisation uses machines to assist workers who still operate and oversee them.

Automation — Machines Replace Humans



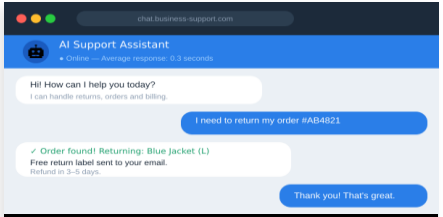
Amazon: 750,000 Kiva robots

Tesla Gigafactory: 1 car/40 secs

- ✔ Computers make ALL decisions — no human needed
- ✔ Works 24/7 — no breaks, no sick days, no wages
- ✔ Examples: Amazon robots, Tesla factory, ATMs, chatbots
- ✖ Very high initial investment in hardware and software
- ✖ Job losses — workers made redundant by machines

Mechanisation — Machines Assist Workers

Ford: workers still operate line



AI chatbot: handles queries 24/7

- ✔ Workers still operate and oversee machinery
- ✔ Less expensive than full automation
- ✔ Examples: conveyor belts, forklift trucks, cash registers
- ✖ Still requires human labour — costs remain
- ✖ Less productive than full automation long-term

CAM & 3D Printing

Computer-Aided Manufacture · Additive Manufacturing

Computer-Aided Manufacture (CAM) & 3D Printing

Technology has fundamentally changed manufacturing. CAM uses computers to control machinery, while 3D printing builds products layer by layer from digital designs.

Computer-Aided Manufacture (CAM)

Computers directly control machines and robots in the production process.

CAM

Computer-Aided Manufacture

robots weld car bodies at Nissan

Nissan's Sunderland plant: robots weld, paint and assemble car bodies using CAM. Workers oversee but machines do the physical work.

- ✅ Consistent quality — no human error in repetitive tasks
- ✅ 24/7 production — machines don't need rest breaks
- ✅ Faster than manual processes — higher output
- ❌ Very high initial investment in equipment
- ❌ Workers need retraining — existing skills may be redundant

3D Printing (Additive Manufacturing)

Builds products layer by layer from a digital design — anything from components to body parts.

3D Printing

Additive manufacturing

NHS prints customised implants

NHS prints customised surgical implants exactly fitting each patient. Airbus prints lightweight aircraft components. Nike prints prototype shoe soles.

- ✅ No mould or tooling needed — designs changed instantly
- ✅ Customised, one-off products at low cost
- ✅ Reduces material waste — only what's needed is used
- ❌ Slow for large-scale production runs
- ❌ Limited materials compared to traditional manufacturing

Exam tip: CAM = computers control machines. 3D printing = building up layers from a digital file. Both reduce labour costs but require high capital investment.

CAM & 3D Printing – Real Machines & Real-World Applications

These are the actual machines and products created by CAM and 3D printing technology — used every day in industry, medicine and design.

CAM – Real Machines in Industry

CNC Robotic Welding Arm
Welds car bodies autonomously

CNC Milling Machine
0.001mm precision engineering

Nissan Sunderland
500,000 cars/yr — UK's best

+ Laser Cutters: clothing · signage · circuit boards · sheet metal — computer-guided precision cuts

3D Printing – Real Applications

NHS Medical Implants
Custom knee/hip implants · dental crowns

Airbus Titanium Parts
1,000+ parts in A350 · 65% lighter

Nike 3D Shoe Soles
Lattice sole · rapid prototyping

+ FDM Desktop Printer: prototyping · education · home use · spare parts — consumer-level 3D printing

Technology in the Service Sector

Contactless · AI · Online services · Apps

Technology in Services – Contactless Payments · AI & Chatbots · Mobile Banking

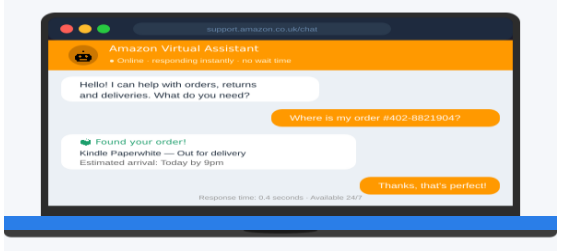
Technology transforms how services are delivered — faster, cheaper and more convenient for customers and businesses.

Contactless Payments

Tap-to-pay with a card or phone. No PIN for low amounts. Speeds up queues, removes cash handling.

- 3 billion UK transactions in 2023
- £100 contactless limit since 2021
- 60% of UK card payments are contactless
- Apple Pay · Google Pay · Visa · Mastercard

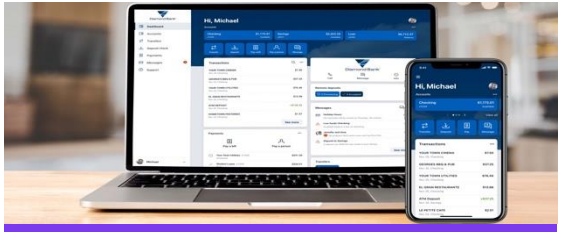
AI & Chatbots



AI handles customer queries, bookings and complaints 24 hours a day without any human staff.

- Amazon Alexa: shopping, queries, home control
- HSBC chatbot: 5 million queries per month
- easyJet bot: rebooking and refunds instantly
- NHS 111 AI: triage reduces A&E pressure

Online & Mobile Banking



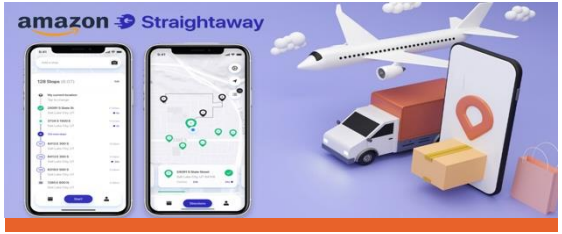
Full bank account management via smartphone app — transfers, payments and loans without ever visiting a branch.

- Monzo: 2m+ UK users · instant alerts · no branches
- Revolut: 40m+ global users · 35+ currencies
- Starling: Best British Bank 5 years running
- All major banks (NatWest, Barclays) now app-first

Technology in Services – E-Commerce · Self-Service Kiosks · AI Personalisation

E-commerce, kiosks and AI personalisation have transformed retail, hospitality and entertainment — giving businesses huge efficiency gains.

E-Commerce & Delivery Apps



Products ordered online, tracked in real time and delivered same or next day using AI logistics and automated warehouses.

- Amazon: 300m products · ships to 200+ countries
- ASOS: 200+countries from UK, zero physical stores
- Deliveroo: AI routes riders to 100,000 restaurants
- Tesco: 20% of grocery sales now online

Self-Service Kiosks

Customers place orders, check in or complete transactions without speaking to any staff — faster service, lower labour costs.

- McDonald's: 50% of orders now placed at kiosks
- Supermarkets: 1 staff serves 4 self-scan checkouts
- Heathrow: self bag-drop eliminates check-in queues
- Hotels: 24/7 check-in kiosks — no reception needed

AI Personalisation

Machine learning analyses your behaviour to predict and suggest exactly what you want — keeping you engaged far longer.

- Spotify: 4 billion AI playlists generated weekly
- Netflix: 80% of viewing comes from AI recommendations
- Amazon: 35% of sales driven by 'Customers also bought'
- TikTok: AI keeps users watching 95 minutes/day average

Technology – Effects on Businesses and Employees

Adopting new technology has major effects on both the business and its workforce. Some are benefits, others create serious challenges.

Benefits for Businesses

- Lower long-term labour costs — fewer staff needed
- Higher and more consistent output quality
- 24/7 production — no breaks, holidays or sick days
- Competitive advantage over slower-adopting rivals

Drawbacks for Businesses

- Very high initial capital investment costs
- Maintenance and upgrades are expensive
- Workers need retraining — costs time and money
- Technology can break down, halting production

Benefits for Employees

- Removes dangerous, dirty or repetitive physical tasks
- Creates new higher-skilled technical jobs
- Workers can focus on creative, problem-solving work

Drawbacks for Employees

- Job losses — machines replace human workers
- Existing skills can become obsolete overnight
- Retraining required — stressful and time-consuming

Exam tip: Always evaluate BOTH sides. A business may gain from automation while the same employees lose their jobs — mention both perspectives.

Key Revision Checklist – 4.2 Technology and Production of Goods and Services

Automation	Machines replace humans entirely — computers make decisions. Amazon robots, Tesla factory, AI chatbots, ATMs
Mechanisation	Machines assist workers who still operate them. Conveyor belts, forklift trucks, cash registers, power tools
CAM	Computers control production machinery. 24/7 consistent quality, no human error. High initial capital cost
CAM examples	CNC robotic arms (welding), CNC milling machines (precision), laser cutters (clothing/circuits), Nissan Sunderland
3D Printing	Builds layer by layer. Customised, no tooling, less waste. Slow for mass production. NHS, Airbus, Nike
3D examples	NHS custom implants, Airbus titanium brackets (1,000+ parts in A350), Nike lattice shoe soles, FDM printers
Service Tech	Contactless payments, AI chatbots, mobile banking, e-commerce, self-service kiosks, AI personalisation (Spotify)
Business FX	Benefits: lower costs, higher output, 24/7. Drawbacks: high investment, maintenance, retraining costs
Employee FX	Benefits: removes dangerous tasks, new jobs. Drawbacks: redundancies, skills obsolete, retraining required